REPORT ON CSEW MENTAL HEALTH TREND ANALYSIS (2019–2024)

WEEK 2 ----- ANA 1

DATASET USED: CSEW MH MODULE – ONS

PREPARED BY: DEEPAK BARANWAL (Data Analyst Intern)

PROJECT NAME: AI MENTAL HEALTH ASSISTANT

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1. Introduction:

Mental health and psychosocial wellbeing became major public health concerns during and after the COVID-19 pandemic. Social stress, domestic issues, abuse-related incidents, and psychological pressure significantly affected individuals across different communities.

This analysis focuses on identifying year-on-year psychosocial and mental health related trends using data extracted from the Crime Survey for England and Wales (CSEW) supplementary tables. The study mainly examines abuse-related prevalence indicators between 2019 and 2024 to understand changes across the COVID-era period.

The objective of this analysis is to identify prevalence trends, observe the impact of the COVID-era period, and visualize changes in mental health related indicators using statistical and graphical analysis techniques.

2. Dataset Description:

The dataset used in this analysis was obtained from the Crime Survey for England and Wales (CSEW) supplementary statistical tables published by the Office for National Statistics (ONS), United Kingdom.

Table S43c from the dataset was selected for analysis because it contains prevalence-based psychosocial indicators associated with domestic abuse, partner abuse, family abuse, threats, force, and sexual assault related measures.

Relevant indicators and yearly prevalence values were extracted for the period 2019–2024. The extracted dataset was then cleaned and transformed into a structured format suitable for trend analysis and Visualization. The final cleaned dataset contained 12 psychosocial and abuse-related indicators with yearly prevalence percentages.

	Indicator	2019-2020	2021-2022	2022-2023	2023-2024
1	Any domestic abuse (partner or family non-phys...	6.063285	5.727713	5.137845	5.35139
2	Any partner abuse (non-physical abuse, threats...	4.477245	3.923697	3.438816	3.553626
3	Any family abuse (non-physical abuse, threats,...	2.136765	2.436163	2.149497	2.122693
4	Partner abuse - non-sexual [note 14]	3.869533	3.22393	2.934945	3.115319
5	Non-physical abuse (emotional, financial) [not...	2.939518	2.548648	2.227238	2.581725
6	Threats or force [note 14]	2.009325	1.493146	1.464506	1.136618
7	Threats [note 14]	1.416103	1.213738	1.050243	0.753369
8	Force [note 14]	1.257405	0.686279	0.771066	0.681944
9	Family abuse - non-sexual [note 14]	1.849323	2.165337	1.899668	1.878842
10	Non-physical abuse (emotional, financial) [not...	1.246234	1.811761	1.401027	1.257299
11	Threats or force [note 14]	0.961539	1.09912	0.864622	0.90785
12	Threats [note 14]	0.74921	0.969015	0.726779	0.697595

3. Data Preprocessing:

Several preprocessing steps were performed before visualization and analysis. The original Excel dataset contained merged cells, extra title rows, and formatting inconsistencies which required cleaning and restructuring.

The relevant sheet (Table S43c) was loaded into Google Colab using the pandas library. Specific rows and yearly prevalence columns were extracted from the dataset to create a structured dataframe.

Unnecessary rows, metadata values, and formatting inconsistencies were removed. The cleaned dataset was then exported into CSV format for further analysis and visualization.

Python libraries including pandas and matplotlib were used for preprocessing and graphical analysis.

4. Trend Analysis:

Trend analysis was performed to study year-wise changes in psychosocial and abuse-related prevalence indicators between 2019 and 2024.

The analysis showed that "Any domestic abuse" had the highest prevalence among the selected indicators. Several indicators demonstrated gradual decline after the COVID-era period, while some remained relatively stable over time.

Partner abuse and threat-related indicators showed noticeable reduction between 2019–2020 and 2023–2024. Family abuse related prevalence remained comparatively stable across multiple years.

The overall trend analysis suggests changing psychosocial conditions and gradual post-pandemic stabilization across several abuse-related indicators.

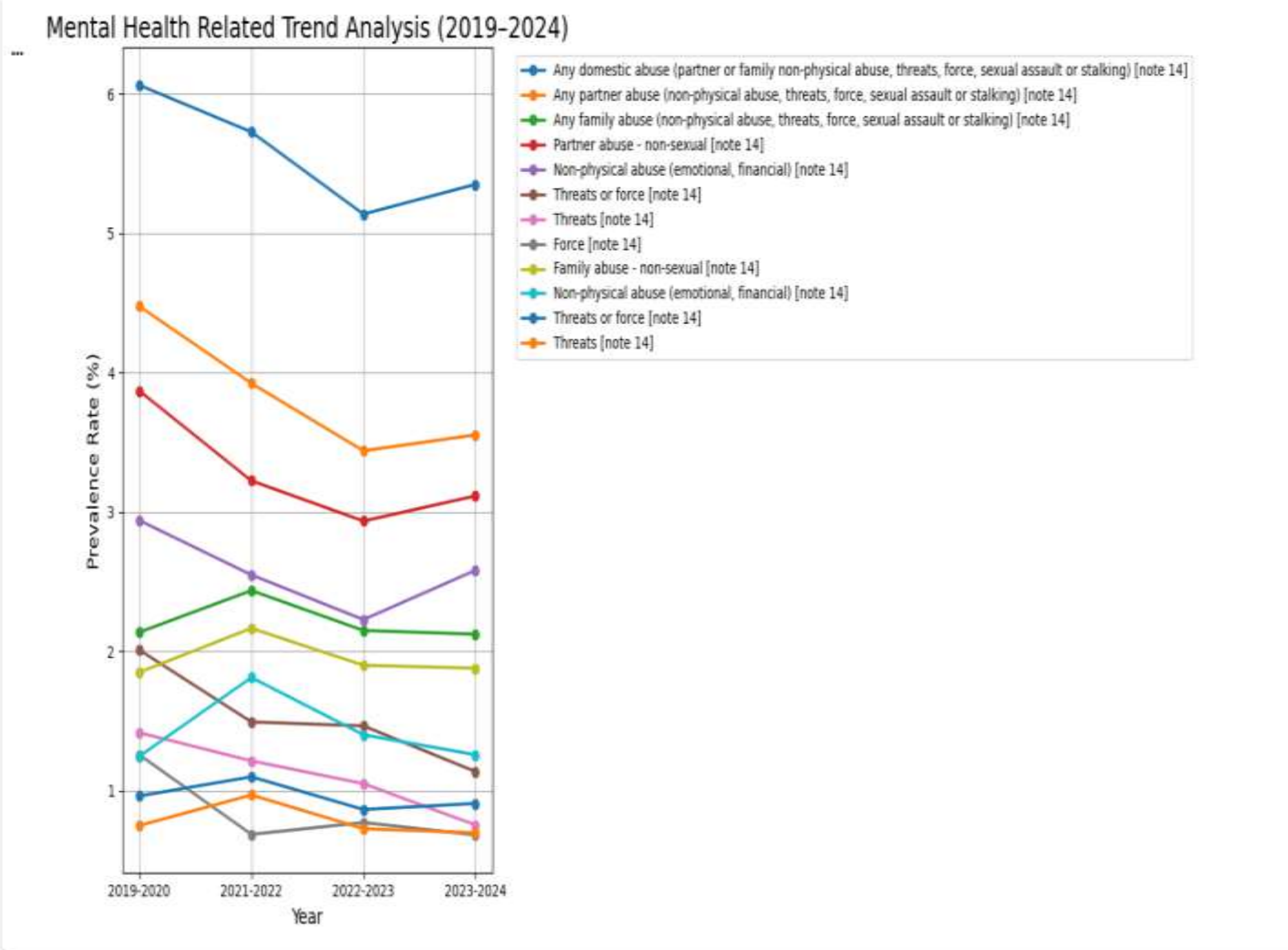
5. COVID-era Analysis:

A dedicated COVID-era trend visualization was created to study the impact of the pandemic period on psychosocial and abuse-related prevalence trends.

The COVID-impact region was highlighted within the visualization to observe prevalence changes during the pandemic years. The analysis indicates that domestic abuse related prevalence remained comparatively elevated during the COVID-era period, reflecting the psychosocial stress and social challenges associated with lockdown conditions and household instability.

Although prevalence values gradually decreased in later years, the visualization demonstrates that the COVID-era period had noticeable influence on psychosocial wellbeing indicators.

MULTI LINE MH TREND CHART



COVID-era highlighted trend analysis for domestic abuse related prevalence indicators.

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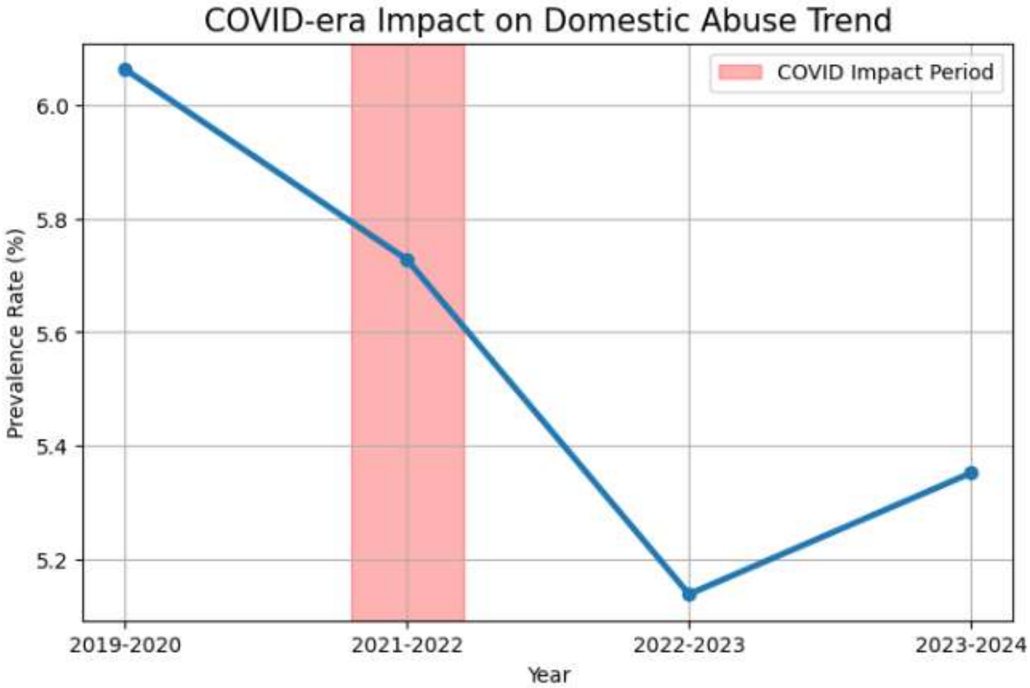

6. Comparative Visualization Analysis:

Additional comparative bar chart visualizations were created to compare prevalence changes between different years and psychosocial indicators.

The bar chart comparison helped identify relative differences between domestic abuse, partner abuse, and family abuse prevalence rates across multiple years.Comparative visualization improved the interpretability of the dataset and helped identify relative prevalence differences more clearly.

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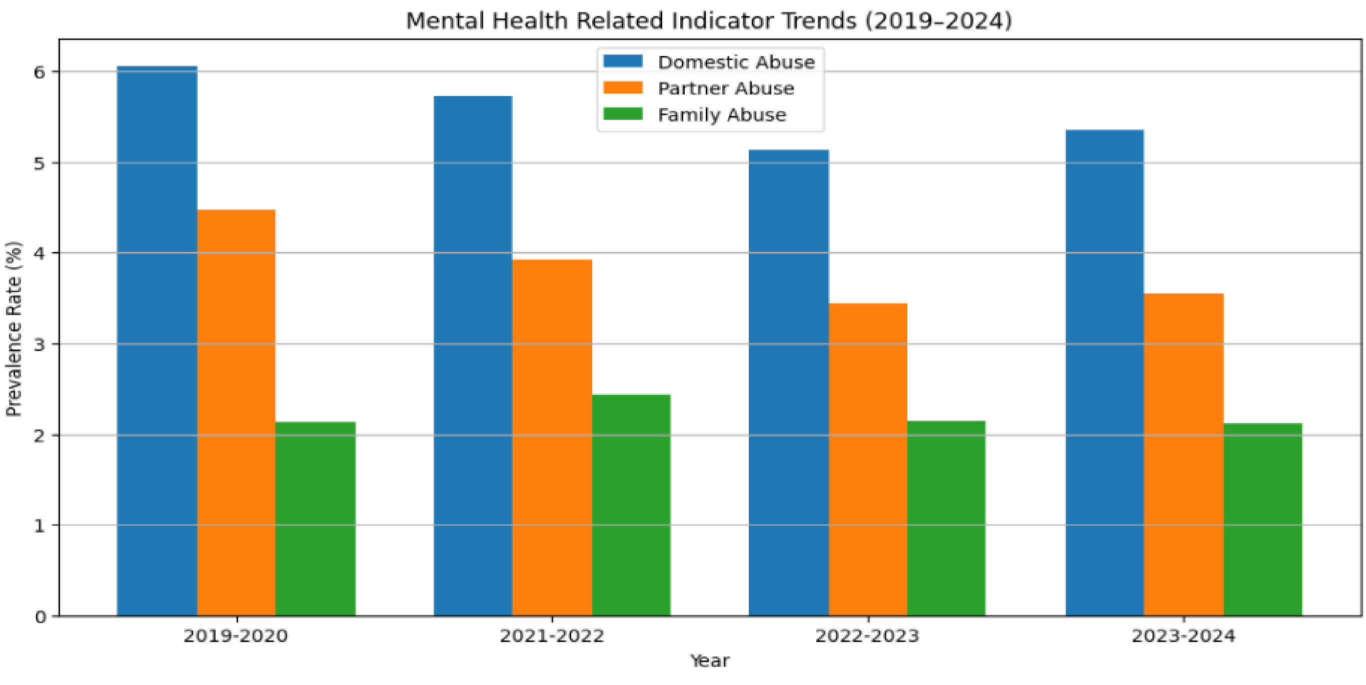

7. Key Insights:

- Domestic abuse related prevalence remained the highest among selected indicators.
- Partner abuse related prevalence showed gradual decline across the observed years.
- Family abuse prevalence remained relatively stable.
- Threat and force related indicators demonstrated noticeable reduction after the COVID-era period.
- COVID-era psychosocial stress may have contributed to elevated prevalence levels during 2020–2022.
- Visualization techniques significantly improved understanding of long-term prevalence trends.

8. Conclusion:

This analysis successfully explored psychosocial and mental health related prevalence trends using the CSEW supplementary dataset from 2019–2024.

The study demonstrated how preprocessing, trend analysis, and visualization techniques can be used to understand changes in psychosocial indicators over time.

The COVID-era analysis highlighted the importance of monitoring public wellbeing and abuse-related prevalence indicators during periods of social disruption and uncertainty.

Overall, the analysis showed that data visualization and trend analysis can support public health understanding, policy interpretation, and stakeholder communication regarding psychosocial wellbeing trends.